

## MICROSOFT POWER BI DATA ANALYST (PL-300) SYLLABUS

**OBJECTIVE:** This Training Helps Business intelligence professionals seeking to elevate their data analysis and visualization expertise. The Training program is structured for data analysts aiming to develop proficiency in utilizing the Power BI Tool for data Analysis

| S.NO | TOPIC                                  | DETAILS                                                                        | DURATION |
|------|----------------------------------------|--------------------------------------------------------------------------------|----------|
| 1    | Get or connect to data                 | <b>Prepare the Data</b>                                                        |          |
|      |                                        | - Identify and connect to data sources or a shared semantic model              |          |
|      |                                        | - Change data source settings, including credentials and privacy levels        |          |
|      |                                        | - Choose between DirectQuery and Import                                        |          |
|      |                                        | - Create and modify parameters                                                 |          |
| 2    | Profile and clean the data             | - Evaluate data, including data statistics and column properties               |          |
|      |                                        | - Resolve inconsistencies, unexpected or null values, and data quality issues  |          |
|      |                                        | - Resolve data import errors                                                   |          |
| 3    | Transform and load the data            | - Select appropriate column data types                                         |          |
|      |                                        | - Create and transform columns                                                 |          |
|      |                                        | - Group and aggregate rows                                                     |          |
|      |                                        | - Pivot, unpivot, and transpose data                                           |          |
|      |                                        | - Convert semi-structured data to a table                                      |          |
|      |                                        | - Create fact tables and dimension tables                                      |          |
|      |                                        | - Identify when to use reference or duplicate queries and the resulting impact |          |
|      |                                        | - Merge and append queries                                                     |          |
|      |                                        | - Identify and create appropriate keys for relationships                       |          |
|      |                                        | - Configure data loading for queries                                           |          |
| 4    | Design and implement a data model      | <b>Model the Data</b>                                                          |          |
|      |                                        | - Configure table and column properties                                        |          |
|      |                                        | - Implement role-playing dimensions                                            |          |
|      |                                        | - Define a relationship's cardinality and cross-filter direction               |          |
|      |                                        | - Create a common date table                                                   |          |
|      |                                        | - Identify use cases for calculated columns and calculated tables              |          |
| 5    | Create model calculations by using DAX | - Create single aggregation measures                                           |          |
|      |                                        | - Use the CALCULATE function                                                   |          |
|      |                                        | - Implement time intelligence measures                                         |          |
|      |                                        | - Use basic statistical functions                                              |          |
|      |                                        | - Create semi-additive measures                                                |          |
|      |                                        | - Create a measure by using quick measures                                     |          |
|      |                                        | - Create calculated tables or columns                                          |          |
|      |                                        | - Create calculation groups                                                    |          |
|      |                                        | - Improve performance by identifying and removing unnecessary rows and columns |          |

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| 6  | Optimize model performance                     | <ul style="list-style-type: none"> <li>- Identify poorly performing measures, relationships, and visuals by using Performance Analyzer and DAX query view</li> <li>- Improve performance by reducing granularity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| 7  | Create reports                                 | <b>Visualize and Analyze the Data</b> <ul style="list-style-type: none"> <li>- Select an appropriate visual</li> <li>- Format and configure visuals</li> <li>- Apply and customize a theme</li> <li>- Apply conditional formatting</li> <li>- Apply slicing and filtering</li> <li>- Configure the report page</li> <li>- Choose when to use a paginated report</li> <li>- Create visual calculations by using DAX</li> </ul>                                                                                                                                                                                                                                                                        |  |
| 8  | Enhance reports for usability and storytelling | <ul style="list-style-type: none"> <li>- Configure bookmarks</li> <li>- Create custom tooltips</li> <li>- Edit and configure interactions between visuals</li> <li>- Configure navigation for a report</li> <li>- Apply sorting to visuals</li> <li>- Configure Sync Slicers</li> <li>- Group and layer visuals by using the selection pane</li> <li>- Drill down into data using interactive visuals</li> <li>- Configure export of report content, and perform an export</li> <li>- Design reports for mobile devices</li> <li>- Enable personalized visuals in a report</li> <li>- Design and configure Power BI reports for accessibility</li> <li>- Configure automatic page refresh</li> </ul> |  |
| 9  | Identify patterns and trends                   | <ul style="list-style-type: none"> <li>- Use the Analyze feature in Power BI</li> <li>- Use grouping, binning, and clustering</li> <li>- Use AI visuals</li> <li>- Use reference lines, error bars, and forecasting</li> <li>- Detect outliers and anomalies</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| 10 | Create and manage workspaces and assets        | <b>Manage and secure Power BI</b> <ul style="list-style-type: none"> <li>- Create and configure a workspace</li> <li>- Configure and update a workspace app</li> <li>- Publish, import, or update items in a workspace</li> <li>- Create dashboards</li> <li>- Choose a distribution method</li> <li>- Configure subscriptions and data alerts</li> <li>- Promote or certify Power BI content</li> <li>- Identify when a gateway is required</li> <li>- Configure a semantic model scheduled refresh</li> </ul>                                                                                                                                                                                      |  |
| 11 | Secure and govern Power BI items               | <ul style="list-style-type: none"> <li>- Assign workspace roles</li> <li>- Configure item-level access</li> <li>- Configure access to semantic models</li> <li>- Implement row-level security roles</li> <li>- Configure row-level security group membership</li> <li>- Apply sensitivity labels</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                          |  |